

Node Kayles on Conic Schreier Graphs: Dihedral and Polyhedral Templates

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Abstract

Let C be a nonsingular conic in $\text{PG}(2, q)$ with q odd. An off-conic point x induces an involution σ_x of $C \cong \mathbb{P}^1(q)$ by exchanging the two points in which each secant through x meets C . In a normal-play capacity-two avoidance game, selected off-conic points forbid the conic points on their pairwise joining lines; if subsequent play is restricted to the conic, the legal continuations form a Node-Kayles position. We prove one reduction and give three applications of it. The reduction holds for any finite involution-generated action: the residual graph is the fixed-point-deleted Schreier graph, and its Grundy value is the xor of the transitive-template numbers weighted only by orbit-multiplicity parity. The applications classify every tame subgroup a legal selected set can induce. Two selected points realize every dihedral group D_{2m} , in either parity of the rotation order, with cycle and path templates and hence Dawson's-chess numbers. Three selected points force the central involution and the even-rotation groups D_{4n} , with empty, K_4 , Möbius-ladder, prism, and ordinary-ladder templates of value in $\{0, 1\}$. The remaining tame possibilities are the finite polyhedral groups S_4 and A_5 . The resulting Grundy classification is an explicit congruence function of q , periodic in q within each torus family, and admits a converse realization theorem. Among admissible prime fields the relative density of P-positions is one half for the even families, and one for every odd-order dihedral group, where the game is always a P-position.

Keywords. Node Kayles; Sprague-Grundy theory; Schreier graph; conic; projective line; dihedral group; $\text{PGL}_2(q)$; Dawson's chess.

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[C308 owed] Keyword list, MSC codes, and the novelty calibration of the abstract are C308 deliverables; the entries above are placeholders inherited from the source manuscript's subject matter.

1 Introduction and main theorem

Node Kayles is the impartial game in which a move at a graph vertex deletes its closed neighbourhood. This paper studies a structured family of Node-Kayles positions arising from a finite-geometric capacity-two game on a conic, and classifies their Grundy values whenever the configuration of selected points induces a tame dihedral or polyhedral subgroup of $\text{PGL}_2(q)$.

The general reduction of late-stage Nofil positions to Node Kayles is known [3]. The correspondence between off-conic points and involutions in $\text{PGL}_2(q)$, and the geometry of triples of such involutions, are also classical; Tranchida [4] recently developed that correspondence in detail for incidence geometries and hypertopes. Our contribution starts after that correspondence. Saturation is detected by fixed points of pair products, so the conic-only residual is a *fixed-point-deleted Schreier graph*; its orbit templates and Grundy values can then be classified completely in the tame case. Tranchida does not consider this residual graph or its Node-Kayles value.

1.1 The reduction and its three applications

The paper has one structural engine and three applications of it. The engine (Theorem 3.1) says that for any finite group G acting on a finite set Ω and generated by a finite set T of involutions, the residual Grundy value depends on Ω only through the parity vector of its orbit multiplicities:

$$\mathcal{G}(R_T(\Omega)) = \bigoplus_{[K]} (m_K \bmod 2) \mathcal{G}(R(G, K, T)), \quad t_K := \mathcal{G}(R(G, K, T)).$$

Geometry therefore enters only by choosing orbit multiplicities; the abstract pair (G, T) chooses a finite table of transitive-template numbers. The three applications compute that table:

- (i) **Two selected points** (Section 4) induce a dihedral group D_{2m} for every m and either parity of the rotation order. The templates are exactly a cycle, a path, or an empty graph, so the numbers are the classical Dawson's-chess values and are unbounded.
- (ii) **Three selected points** (Section 5) must contain the central involution, which forces the even-rotation groups D_{4n} together with the Klein-four boundary. The templates are exactly an empty graph, K_4 , a Möbius ladder, a prism, or an ordinary ladder, all of value in $\{0, 1\}$.
- (iii) **Polyhedral triples** (Section 6) are the remaining tame possibilities, S_4 and A_5 ; A_4 cannot occur. This is a finite boundary and is settled by exact computation of the templates.

Section 7 converts these tables into closed field formulas, P/N congruences, and a density theorem. Section 8 collects the converse realization statements and states the boundaries of the classification.

1.2 Conventions and standing hypotheses

Throughout, q is odd, $C \subset \text{PG}(2, q)$ is a nonsingular conic, and

$$p = \text{char } \mathbb{F}_q.$$

We work in the *tame* case, meaning that p does not divide the order of the subgroup under consideration. We write

$$D_{2m} = \langle r, s : r^m = s^2 = 1, srs = r^{-1} \rangle,$$

so that D_{2m} has order $2m$. The pair family below is indexed by D_{2m} and the triple family by D_{4n} with $m = 2n$; the two indices are kept distinct at every theorem boundary.

All game values refer to the continuation in which only conic points may be played. They need not equal the value of a larger position in which further off-conic moves remain available.

1.3 Main theorem

Theorem 1.1 (Classification, tame dihedral configurations). *Let q be odd and let S be a legal set of selected off-conic points whose induced involutions generate a tame dihedral subgroup $G \leq \text{PGL}_2(q)$. Then $|S| \in \{2, 3\}$, and:*

- (a) every transitive residual template of (G, T) is an empty graph, K_4 , a cycle C_{2m} , a path P_m , an ordinary ladder L_k , a prism $C_{2n} \square K_2$, or a Möbius ladder M_{4n} ;
- (b) the conic-only residual Grundy value is the xor of those template numbers weighted by the parity of their orbit multiplicities, and is an explicit congruence function of q , periodic in q within each torus family;
- (c) every template in (a) is realized over infinitely many prime fields.

The two-point family realizes every m and has unbounded values; the three-point family realizes exactly the even rotation orders and has values in $\{0, 1\}$.

The statement is proved in pieces: (a) in Sections 4 and 5, (b) in Section 7, and (c) in Section 8.

[C307 owed] Theorem 1.1 is stated here at the strength the migrated source manuscript proves. C307 extends it to the polyhedral families using the C284 completeness theorem, the C289 ρ refinement, and the C290 congruence laws, and must name the three exclusions (off-conic continuation, wild characteristic, growing full/subfield $\text{PSL}_2/\text{PGL}_2$ groups) in the same breath, per Fable's answer (10).

2 From conic saturation to Node Kayles

Choose coordinates in which

$$C : XZ = Y^2,$$

and parameterize

$$c(t) = [t^2 : t : 1], \quad t \in \mathbb{F}_q,$$

with $c(\infty) = [1 : 0 : 0]$.

For an off-conic point x , projection from x exchanges the two conic points on each secant through x . Denote the resulting involution of C by σ_x . For $x = [a : b : c]$ this involution is represented by the traceless matrix

$$M_x = \begin{pmatrix} b & -a \\ c & -b \end{pmatrix},$$

acting on $\mathbb{P}^1(q)$ as

$$\sigma_x(t) = \frac{bt - a}{ct - b}. \quad (1)$$

Let S be a legal set of selected off-conic points. A conic point is *dead* when it lies on a line through two points of S .

Lemma 2.1 (Pair-product fixed points are exactly the dead conic points). *For distinct selected points x, y and $u \in C$,*

$$u \in xy \iff (\sigma_x \sigma_y)(u) = u. \quad (2)$$

Proof. Let $v = \sigma_y(u)$. The points u, v, y are collinear. If $\sigma_x(v) = u$, then u, v, x are also collinear. When $u \neq v$, the line through u, v is unique, so it contains x and y , and hence $u \in xy$. When $u = v$, both x and y lie on the unique tangent to C at u , so again $u \in xy$.

Conversely, suppose $u \in xy$. Let v be the other point of $C \cap xy$, counted with multiplicity. Projection from y exchanges u and v , and projection from x exchanges v and u . Thus $\sigma_x \sigma_y(u) = u$. $\square$

Define

$$D_S = \bigcup_{\{x,y\} \subset S} \text{Fix}(\sigma_x \sigma_y). \quad (3)$$

On the live vertex set $C \setminus D_S$, join u to $\sigma_x(u)$ for every $x \in S$, suppressing loops and repeated edges. Call the resulting graph R_S .

Theorem 2.2 (Exact residual reduction). *A legal conic move at $u \in C \setminus D_S$ deletes exactly the closed neighbourhood $N_{R_S}[u]$. Hence play restricted to the conic is Node Kayles on R_S .*

Proof. After selecting u , a live conic point $v \neq u$ becomes dead exactly when u, v, x are collinear for some $x \in S$. By definition of the projection involution this is equivalent to $v = \sigma_x(u)$. Thus the newly unavailable conic points are precisely u and its graph neighbours. $\square$

The graph is static throughout the conic-only continuation. A line through two selected conic points contains no third conic point, while a line through a newly selected conic point and an old off-conic point $x \in S$ removes exactly its projection partner. Later moves therefore pass to induced subgraphs of the original R_S and never add edges. This is what makes the reduction genuine Node Kayles rather than a dynamically changing graph game.

3 Orbit templates and mod-two cancellation

Let a finite group G act on a finite set Ω , and let T be a finite set of involutions generating G . Define

$$D_T(\Omega) = \bigcup_{\{\sigma, \tau\} \in \binom{T}{2}} \text{Fix}_\Omega(\sigma\tau), \quad (4)$$

where the unordered-pair notation is unambiguous because $\tau\sigma = (\sigma\tau)^{-1}$ has the same fixed points as $\sigma\tau$. Equivalently, for involutions,

$$\sigma\tau(x) = x \iff \sigma(x) = \tau(x),$$

which is the common-projection-partner condition of Lemma 2.1. Define also

$$R_T(\Omega) = \text{Sch}(G, \Omega, T)[\Omega \setminus D_T(\Omega)], \quad (5)$$

with loops and repeated edges suppressed. For a subgroup $K \leq G$ write

$$R(G, K, T) = R_T(G/K). \quad (6)$$

Theorem 3.1 (Orbit-template reduction). *If $\Omega \cong \bigsqcup_{[K]} m_K(G/K)$, then Sprague-Grundy additivity gives*

$$\mathcal{G}(R_T(\Omega)) = \bigoplus_{[K]} (m_K \bmod 2) \mathcal{G}(R(G, K, T)). \quad (7)$$

Proof. Every generator preserves each G -orbit, so both the Schreier edges and the pair-product fixed points split orbit by orbit. Although $D_T(\Omega)$ need not itself be G -stable when T is not closed under conjugation, every G -equivariant bijection commutes with every element of T ; it therefore transports D_T and the residual graph. Thus each orbit of type G/K contributes the well-defined template $R(G, K, T)$. Grundy values xor over disjoint graph components, and two equal numbers cancel, giving (7). $\square$

Consequently, once the finite table $t_K = \mathcal{G}(R(G, K, T))$ is known, the ambient action enters only through the parity vector $(m_K \bmod 2)_{[K]}$. The rest of the paper computes these tables.

3.1 Burnside reformulation

Fix G and T as above and define $\Phi_T(\Omega) = \mathcal{G}(R_T(\Omega))$ for every finite G -set Ω .

Proposition 3.2. *The map Φ_T is additive under disjoint union and extends uniquely to a homomorphism from the additive group underlying the Burnside ring,*

$$\Phi_T : A(G) \longrightarrow (\mathbb{N}_0, \oplus). \quad (8)$$

It vanishes on $2A(G)$ and therefore factors through the mod-two Burnside group $A(G) \otimes_{\mathbb{Z}} \mathbb{F}_2$.

Proof. The construction R_T respects disjoint unions, and Grundy values xor over graph components, so $\Phi_T(\Omega \sqcup \Omega') = \Phi_T(\Omega) \oplus \Phi_T(\Omega')$. The Grothendieck-group universal property gives the extension. Finally $\Phi_T(\Omega \sqcup \Omega) = 0$, so doubled classes lie in the kernel. $\square$

Corollary 3.3 (Bulk cancellation). *If a finite G -set has f free orbits and exceptional part Ω_{exc} , then*

$$\Phi_T(\Omega) = (f \bmod 2) \Phi_T(G/1) \oplus \Phi_T(\Omega_{\text{exc}}). \quad (9)$$

Arbitrarily many generic free orbits therefore contribute either nothing or a single parity bit.

Remark 3.4. This is a reformulation of Theorem 3.1, not an additional source of computational information: the additive group of $A(G)$ is already free on the transitive G -sets, and no compatibility with Burnside-ring multiplication is asserted.

[C308 owed] Proposition 3.2 and Corollary 3.3 are the Lean-anchored results of this section (C262, Burnside.lean). C308 attaches the formal anchor citation to the corollary, per Fable’s answer (3).

4 Two selected points: every dihedral order

4.1 Legal dihedral configurations

Lemma 4.1 (Collinearity of reflection centres). *For any dihedral subgroup of $\text{PGL}_2(q)$, regardless of the parity of its rotation order, all reflection centres lie on one projective line.*

Proof. Over the splitting field a dihedral normal form represents every reflection by a matrix proportional to

$$\begin{pmatrix} 0 & \lambda \\ 1 & 0 \end{pmatrix},$$

whose centre has coordinate $b = 0$. The line spanned by the reflection centres is Galois invariant and therefore descends to $\mathbb{F}_q$. $\square$

A legal selected set may have any size, so we record the complete taxonomy of legal sets whose induced involutions generate a dihedral group. By Lemma 4.1 a legal set contains at most two reflections, and it contains at most one central involution z . Hence a legal dihedral configuration has either two or three selected points:

selected involutions	generated group	admissible m
two reflections	D_{2m}	every $m \geq 3$, both parities
one reflection and the central z	V_4	$m = 2$ boundary
two reflections and the central z	D_{4n} , $m = 2n$	even rotation order only

Table 1: Legal dihedral configurations. A set of four or more selected points cannot induce a dihedral group, since it would contain three collinear reflection centres.

The third row is the group D_{4n} analysed in Section 5. The first row is the general dihedral family: any two selected off-conic points x, y induce involutions σ_x, σ_y generating

$$G = \langle \sigma_x, \sigma_y \rangle \cong D_{2m}, \quad m = \text{ord}(\sigma_x \sigma_y),$$

with no central involution required, so this family exists for *every* m , including the odd rotation orders that admit no legal triple.

Throughout this section $\{x, y\}$ is a legal pair, $G \cong D_{2m}$ is tame ($p \nmid 2m$), and the generating involution set is

$$T = \{\sigma_x, \sigma_y\} = \{s, sr^d\}, \quad \gcd(d, m) = 1.$$

Since $\gcd(d, m) = 1$ we may replace r by r^d and take $d = 1$, so $T = \{s, sr\}$ and $\sigma_x \sigma_y = r$.

4.2 Point stabilizers and templates

As in Proposition 5.6 below, every tame point stabilizer of G on $\mathbb{P}^1(q)$ is cyclic, hence conjugate to 1, to the rotation subgroup $\langle r \rangle$, or to a reflection subgroup $\langle sr^e \rangle$. The reflections form one conjugacy class when m is odd and two classes, represented by $\langle s \rangle$ and $\langle sr \rangle$, when m is even.

Theorem 4.2 (Two-generator templates). *For $T = \{s, sr\}$,*

$$R(D_{2m}, 1, T) \cong C_{2m}, \quad R(D_{2m}, \langle sr^e \rangle, T) \cong P_m, \quad R(D_{2m}, \langle r \rangle, T) = \emptyset, \quad (10)$$

where C_{2m} is the cycle on $2m$ vertices and P_m the path on m vertices.

Proof. Free orbit. The regular action has no deleted vertices, so the template is the Cayley graph $\text{Cay}(G, T)$. It is 2-regular, and its edges alternate between the two involution generators. Since T generates G , that alternation visits all $2m$ group elements in a single closed walk. Hence it is one $2m$ -cycle.

Rotation orbit. The two cosets of $\langle r \rangle$ are each fixed by $\sigma_x \sigma_y = r \in \langle r \rangle$, so both are deleted by (4) and the template is empty.

Reflection orbit. Label the cosets $D_{2m}/\langle s \rangle$ by $j \in \mathbb{Z}/m$ via $j \leftrightarrow r^j \langle s \rangle$. Left multiplication gives

$$s : j \mapsto -j, \quad sr : j \mapsto -1 - j, \quad \text{product } r : j \mapsto j + 1. \quad (11)$$

The deleted vertices are $\text{Fix}(\sigma_x \sigma_y) = \text{Fix}(r)$ on this coset set; but $r : j \mapsto j + 1$ is fixed-point-free on $\mathbb{Z}/m$, so no vertex of a reflection orbit is deleted. The two reflection maps in (11) have product the m -cycle r , so their edge union is connected of maximum degree two. Their fixed points, which produce suppressed loops rather than edges, are the solutions of $2j \equiv 0$ and $2j \equiv -1 \pmod{m}$; counting solutions in each parity of m leaves exactly two vertices of degree one and $m - 2$ of degree two. A connected graph of maximum degree two with exactly two degree-one vertices is the path P_m . For a general reflection stabilizer $\langle sr^e \rangle$, the automorphism $r \mapsto r, s \mapsto sr^e$ of D_{2m} carries $\langle s \rangle$ to $\langle sr^e \rangle$ and $T = \{s, sr\}$ to another two-reflection set with product r , so the same computation applies and the template is P_m independently of e . $\square$

4.3 Template numbers

Node Kayles on a path is Dawson's chess: deleting a closed neighbourhood removes an interior vertex together with its at most two neighbours, splitting P_m into two shorter paths, which is the move rule of the octal game 0.137 [8]. Its Grundy sequence

$$\mathcal{G}(P_m) = \text{A002187}(m) = 0, 1, 1, 2, 0, 3, 1, 1, 0, 3, 3, 2, 2, 4, 0, 5, \dots \quad (m = 0, 1, 2, \dots) \quad (12)$$

is eventually periodic with period 34 [9].

For the cycle, every vertex of C_k has the same closed neighbourhood shape, and deleting it removes three consecutive vertices, leaving P_{k-3} . Vertex-transitivity makes this the only option up to isomorphism, so

$$\mathcal{G}(C_k) = \text{mex}\{\mathcal{G}(P_{k-3})\} \quad (k \geq 3). \quad (13)$$

Lemma 4.3 (Free templates are neutral). *For every $m \geq 2$, $\mathcal{G}(C_{2m}) = 0$.*

Proof. By (13), $\mathcal{G}(C_{2m}) = \text{mex}\{\mathcal{G}(P_{2m-3})\}$, which is zero unless $\mathcal{G}(P_{2m-3}) = 0$. The index $2m - 3$ is odd, so it suffices to know that the Dawson sequence (12) has no zero at an odd index. Its period is 34, which is even, so the index parity is constant along each residue class modulo the period; hence it suffices to check the finitely many indices up to the end of the first full period together with the pre-periodic part, and there every zero of A002187 occurs at an even index. Therefore $\mathcal{G}(P_{2m-3}) \neq 0$ and the mex is zero. $\square$

Free orbits therefore never contribute to the value of a two-point residual.

Theorem 4.4 (Pair orbit formula). *Let Ω be a tame D_{2m} -set with f free orbits and ρ reflection orbits. Then*

$$\mathcal{G}(R_T(\Omega)) = (f \bmod 2) \mathcal{G}(C_{2m}) \oplus (\rho \bmod 2) \mathcal{G}(P_m) = (\rho \bmod 2) \mathcal{G}(P_m). \quad (14)$$

Proof. Theorem 3.1 gives the first equality, with the rotation orbits contributing the empty template. Lemma 4.3 kills the free term. $\square$

Corollary 4.5 (Period 34 in the rotation order). *For a fixed reflection-orbit parity, the two-point residual value $(\rho \bmod 2) \mathcal{G}(P_m)$ is eventually periodic in m with period 34.*

Proof. Immediate from (14) and the eventual period-34 periodicity of (12). $\square$

The Dawson zeros $\mathcal{G}(P_m) = 0$ at even $m \in \{4, 8, 14, 20, 24, 28, 34, \dots\}$ give infinite families of even-order dihedral groups whose two-point game is always a P-position. This is the two-generator analogue of the always-deleted rotation torus, but arises from a nontrivial path cancellation rather than from a deletion.

5 Three selected points: the dihedral ladder family

5.1 The Klein-four boundary

The smallest dihedral boundary is $D_4 \cong V_4$. A generating involution triple is necessarily the set of all three nonidentity elements.

Theorem 5.1 (Klein-four boundary). *Let a legal tame conic-involution triple generate $G \cong V_4$, and let s be the number of its three involutions having rational fixed points on C . The realizable split counts satisfy*

$$s \in \{1, 3\} \ (q \equiv 1 \bmod 4), \quad s \in \{0, 2\} \ (q \equiv 3 \bmod 4). \quad (15)$$

The dead set has size $2s$, and the residual graph is

$$\frac{q+1-2s}{4} K_4, \quad (16)$$

a coefficient denoting a disjoint union of that many copies. Consequently

$$\mathcal{G}(R_T) = \frac{q+1-2s}{4} \bmod 2. \quad (17)$$

Proof. The product of any two nonidentity elements of V_4 is the third, so (4) deletes exactly the union of the fixed-point sets of the three involutions. A tame point stabilizer in $\mathrm{PGL}_2(q)$ is cyclic, so two distinct nonidentity elements of V_4 cannot fix the same conic point. Each split involution has two rational fixed points and each nonsplit involution has none, giving $|D_T| = 2s$.

The split-count restriction is the parity law for a tame Klein four subgroup of $\mathrm{PGL}_2(q)$: the number of split nonidentity elements has parity opposite to $(q-1)/2$, giving exactly the alternatives in (15). Equivalently the orbit equation $q+1 = 2s+4f$ forces the same parity, and the standard split/nonsplit normal forms realize both listed values. In particular divisibility of $q+1-2s$ by four is automatic.

The V_4 -action on the live points is free: a nontrivial stabilizer would again contain one of the involutions whose fixed points were deleted. Every live orbit therefore has four points, and on such an orbit the three involutions give the three perfect matchings of K_4 . Since one Node-Kayles move deletes a whole K_4 , each orbit has Grundy value one, and disjoint-union additivity gives (17). $\square$

In the graph notation below this is the $n=1$ boundary $M_4 = K_4$, while the reflection-stabilized template is $L_0 = \emptyset$. It is treated separately because the T_d parametrization and the pendant-ladder proof for $n \geq 2$ degenerate at $n=1$.

5.2 The legal triples T_d

No dihedral group whose rotation subgroup has odd order can arise from a legal generating involution triple: all of its involutions are reflections, and their centres are collinear by Lemma 4.1. For the remainder of the section fix $n \geq 2$ and suppose

$$G \cong D_{4n} = \langle r, s : r^{2n} = s^2 = 1, srs = r^{-1} \rangle, \quad z = r^n.$$

The noncentral involutions are the reflections sr^e , so three noncentral involution centres are collinear and cannot be a legal selected triple. A legal generating involution triple must therefore contain z and two reflections. After rechoosing the reflection generator and interchanging the reflections, it has the form

$$T_d = \{z, s, sr^d\}. \quad (18)$$

Proposition 5.2. *The triple T_d generates D_{4n} exactly when $\gcd(d, n) = 1$.*

Proof. The subgroup contains $s(sr^d) = r^d$ and r^n , so its rotation subgroup is $\langle r^d, r^n \rangle = \langle r^{\gcd(d, n)} \rangle$. This equals $\langle r \rangle$ precisely when $\gcd(d, n) = 1$. $\square$

Under an automorphism $r \mapsto r^u$, $s \mapsto r^v s$ with $u \in (\mathbb{Z}/2n)^\times$, the parameter d changes to $\pm ud$. Therefore if n is even every admissible d is odd and there is one automorphism class, while if n is odd there are two classes, represented by odd and even d . The central centre is not on the reflection-centre line, so every generating triple described above is geometrically legal.

5.3 Free-orbit templates

Let M_{4n} denote the graph obtained from C_{4n} by joining antipodal vertices, and let $L_k = P_k \square K_2$ denote the ladder with k rungs.

Theorem 5.3. *For the regular action,*

$$R(D_{4n}, 1, T_d) \cong \begin{cases} M_{4n}, & d \text{ odd}, \\ C_{2n} \square K_2, & d \text{ even}. \end{cases} \quad (19)$$

The even case occurs only when n is odd.

Proof. No nonidentity element fixes a point in the regular action, so no vertices are deleted.

The two reflection generators alternate along the cosets of $\langle r^d \rangle$. If d is odd then $\gcd(d, 2n) = 1$, so their edges form one $4n$ -cycle. Multiplication by $z = r^n$ joins opposite vertices: centrality of z makes this chord matching invariant under the reflection cycle, and after two reflection steps one advances by d , so z advances by exactly half of that $4n$ -cycle. This gives M_{4n} .

If d is even then n is odd and $\gcd(d, 2n) = 2$. The reflection edges form two $2n$ -cycles, and z joins corresponding vertices of the two cycles. This is $C_{2n} \square K_2$. $\square$

Theorem 5.4. *For every $n \geq 2$, $\mathcal{G}(M_{4n}) = 1$.*

Proof. The graph is vertex-transitive. After a move at v , the graph $M_{4n} - N[v]$ is the opposite-end-pendant ladder on $2n - 3$ rungs. To see this explicitly, label M_{4n} by $\mathbb{Z}/(4n)$, with cycle edges $i \sim i + 1$ and chords $i \sim i + 2n$, and take $v = 0$. Removing $N[0] = \{0, 1, 2n, 4n - 1\}$ leaves the paired vertices

$$2 + i \longleftrightarrow 2n + 2 + i, \quad 0 \leq i \leq 2n - 4,$$

as the $2n - 3$ rungs. The unpaired vertex $2n - 1$ is pendant at the right end of the first rail, while $2n + 1$ is pendant at the left end of the second rail. This is exactly Brown et al.'s third ladder variation $-L_{(2n-3) \times 2}^- [1]$.

Their ladder theorem states $\mathcal{G}(-L_{k \times 2}^-) = 0$ for every integer $k \geq 1$. Here $k = 2n - 3$ is odd and at least one. Their $k = 1$ base graph is P_4 , with value zero, and $k = 3$ is already covered by the same theorem; the first two values arising here are therefore included explicitly, not inferred from an eventual pattern. Hence every option of M_{4n} has value zero, and $\mathcal{G}(M_{4n}) = \text{mex}\{0\} = 1$. $\square$

Theorem 5.5. *For every $k \geq 3$, $\mathcal{G}(C_k \square K_2) = 0$.*

Brown et al.'s prism corollary [1] states this for every $k \geq 3$, with no parity or residue restriction. In particular it applies to the only prisms used here, where $k = 2n \geq 6$ and n is odd.

5.4 Point stabilizers in the tame projective action

Let $A = \langle r \rangle \cong \mathbb{Z}/2n$ and $K_e = \langle sr^e \rangle$.

Proposition 5.6. *Every point stabilizer of D_{4n} on $\mathbb{P}^1(q)$ is conjugate to one of*

$$1, \quad A, \quad K_0, \quad K_1. \quad (20)$$

Proof. A point stabilizer in $\text{PGL}_2(q)$ is contained in a Borel subgroup. Its normal unipotent subgroup has order a power of p ; since $p \nmid |G|$, the intersection of G with that subgroup is trivial. Projection to the cyclic quotient of the Borel by its unipotent subgroup is therefore injective on the point stabilizer, so the stabilizer is cyclic.

If the stabilizer contains a nonidentity rotation, the point is one of the two fixed points of the rotation torus over the splitting field and is therefore fixed by all of A . An apparent

intermediate rotation stabilizer $\langle r^k \rangle$ immediately enlarges to A , and it cannot enlarge further inside D_{4n} , since adjoining any reflection to A would make the stabilizer dihedral and hence noncyclic. Otherwise the stabilizer is generated by a reflection. Because the rotation order is even, the reflections form two conjugacy classes, represented by K_0 and K_1 . $\square$

Proposition 5.7. *The rotation-stabilized template is empty: $R(D_{4n}, A, T_d) = \emptyset$.*

Proof. The two cosets of A are fixed by $s(sr^d) = r^d \in A$, so both are deleted by (4). $\square$

5.5 Reflection-stabilized templates

Label the cosets D_{4n}/K_e by $j \in \mathbb{Z}/(2n)$ via $j \leftrightarrow r^j K_e$. The generators act as

$$z : j \mapsto j + n, \quad s : j \mapsto e - j, \quad sr^d : j \mapsto e - d - j. \quad (21)$$

The two reflection maps have product $j \mapsto j + d$. The deleted vertices are the solutions of

$$2j = e + n \quad \text{or} \quad 2j = e - d + n \pmod{2n}. \quad (22)$$

Lemma 5.8 (A double cover of a tree is untwisted). *Let a fixed-point-free involution z act on a graph X . Suppose a z -invariant edge set E_0 descends to a graph $Y = X/\langle z \rangle$, with every edge of Y having exactly two z -paired lifts. If Y is a tree, then $(V(X), E_0)$ is the disjoint union of two copies of Y , exchanged by z . After adding every edge $v \sim z(v)$, the result is $Y \square K_2$.*

Proof. Choose a root of Y and one of its two lifts. Every vertex of a tree has a unique path from the root, so lifting that path selects a unique lift of every vertex and produces one copy of Y . Applying z produces the other copy. There is no monodromy because the tree has no cycle. The added z -edges join the two lifts of each quotient vertex, which are exactly the K_2 -fibres of the Cartesian product. $\square$

Theorem 5.9. *For odd d ,*

$$R(D_{4n}, K_e, T_d) \cong L_{n-1}, \quad (23)$$

independently of e . For even d , necessarily n odd,

$$R(D_{4n}, K_e, T_d) \cong \begin{cases} L_n, & e \text{ even}, \\ L_{n-2}, & e \text{ odd}. \end{cases} \quad (24)$$

Proof. Because $\gcd(d, n) = 1$, reduction modulo n makes translation by d transitive on $\mathbb{Z}/n$, regardless of the parity of d . Quotient the coset set by the fixed-point-free central involution $z : j \mapsto j + n$. On the quotient, the two reflection maps are involutions whose product is this transitive translation. Their simple edge union is connected and has maximum degree two. The two quotient reflections have two fixed points in total: each has one when n is odd, while for even n one has two and the other has none. Suppressing the corresponding loops leaves $n - 1$ edges, so the quotient reflection graph is the path P_n , with those fixed points as its endpoints.

Each solvable congruence in (22) has two solutions differing by n , hence deletes one z -orbit. After reduction modulo n , the two congruences are exactly the fixed-point equations for the two quotient reflections, so every deleted z -orbit is an endpoint of P_n . If d is odd, the right-hand sides in (22) have opposite parity, so exactly one endpoint orbit is deleted and the live quotient is P_{n-1} . If d is even, then n is odd; for even e neither congruence is solvable and the live quotient is P_n , while for odd e both are solvable and the live quotient is P_{n-2} .

The reflection edges above quotient path edges satisfy Lemma 5.8, and the z -edges are its fibre edges. A reflection above a suppressed endpoint loop either gives loops upstairs or duplicates its z -edge, neither of which changes the underlying simple graph. The three live quotients therefore lift to L_{n-1} , L_n , and L_{n-2} respectively. $\square$

Brown et al. [1] prove

$$\mathcal{G}(L_k) = \begin{cases} 1, & k \text{ odd}, \\ 0, & k \text{ even}, \end{cases} \quad (25)$$

and therefore

$$\mathcal{G}(R(D_{4n}, K_e, T_d)) = \begin{cases} \mathbf{1}_{2|n}, & d \text{ odd}, \\ 1, & d \text{ even}. \end{cases} \quad (26)$$

5.6 Complete orbit formulas

Let Ω be a finite D_{4n} -set all of whose point stabilizers are conjugate to one of $1, A, K_0, K_1$; by Proposition 5.6 the natural tame action on $\mathbb{P}^1(q)$ has this property. Write

$$\Omega \cong f(D_{4n}/1) \sqcup b_A(D_{4n}/A) \sqcup b_0(D_{4n}/K_0) \sqcup b_1(D_{4n}/K_1) \quad (27)$$

with nonnegative integer multiplicities.

Theorem 5.10 (Triple orbit formula). *For odd d ,*

$$\mathcal{G}(R_{T_d}(\Omega)) = (f \bmod 2) \oplus (\mathbf{1}_{2|n}((b_0 + b_1) \bmod 2)). \quad (28)$$

For even d , necessarily n odd,

$$\mathcal{G}(R_{T_d}(\Omega)) = (b_0 + b_1) \bmod 2. \quad (29)$$

The multiplicity b_A does not occur because its template is empty.

For the natural action on $\mathbb{P}^1(q)$, each exceptional class occurs at most once and the exceptional point sets are pairwise disjoint: otherwise one point stabilizer would contain either two distinct reflections or a reflection and a nonidentity rotation, contradicting the cyclicity conclusion of Proposition 5.6. (Here $2n \geq 4$, so the two reflection classes and the central rotation are genuinely distinct.) Writing the orbit indicators as $\varepsilon, a_0, a_1 \in \{0, 1\}$,

$$q + 1 = 4nf + 2\varepsilon + 2n(a_0 + a_1). \quad (30)$$

The complete transitive-template table for the dihedral configurations, including the Klein-four boundary, is therefore:

group and triple	stabilizer	residual template	$\mathcal{G}$
V_4 , all three involutions	1	K_4	1
V_4 , all three involutions	nontrivial cyclic	$\emptyset$	0
D_{4n} , odd d	1	M_{4n}	1
D_{4n} , even d	1	$C_{2n} \square K_2$	0
D_{4n} , any d	A	$\emptyset$	0
D_{4n} , odd d	K_e	L_{n-1}	$\mathbf{1}_{2 n}$
D_{4n} , even d , even e	K_e	L_n	1
D_{4n} , even d , odd e	K_e	L_{n-2}	1
D_{2m} , two reflections	1	C_{2m}	0
D_{2m} , two reflections	$\langle sr^e \rangle$	P_m	A002187(m)
D_{2m} , two reflections	$\langle r \rangle$	$\emptyset$	0

Table 2: The complete transitive-template table for legal tame dihedral configurations. The graph families of Theorem 1.1 are a corollary of this one table; the ambient projective line merely supplies its orbit multiplicities.

6 Polyhedral triples: the finite boundary

Beyond the dihedral configurations the tame possibilities are finite. The next subgroup in the usual polyhedral list, A_4 , cannot occur as a generated group here: its only involutions are the three nonidentity elements of its normal Klein four subgroup, so any subgroup of A_4 generated by involutions is contained in that V_4 , not all of A_4 . The remaining polyhedral possibilities are S_4 and A_5 .

[C307 owed] This section is the structural placeholder for the promoted polyhedral content. Per Fable's change 4 it must *lead with completeness* — the C284 classification theorem that all admissible S_4/A_5 coset templates are tabulated, with A_4 impossible — and then present the A_5 (3, 5, 5; $\rho = 3/5$) split as the load-bearing refinement, proved as an $\text{Aut}(G)$ -orbit invariant (C289) with no stronger geometric interpretation. The ten-row (σ, ρ , class size, $t_1 \dots t_5$) table belongs here in the body; edge lists and replay counts go to Appendix A (Fable's answer 4). C289's mirror lemma upgrades six $t_1 = 0$ entries from computed to proved; the $\rho = 3$ regular zero stays computational and must be visibly marked as such. The free-orbit table below is the only polyhedral material the source manuscript carries and is retained here unchanged pending that integration.

6.1 Regular polyhedral templates

For a regular action there are no deleted vertices, so the template is the cubic Cayley graph $\text{Cay}(G, T)$. Generating involution triples are classified by the sorted orders of their three pairwise products. Exact Node-Kayles computation gives Table 3.

G	pair-product-order class	graph	$\mathcal{G}$
S_4	(2, 3, 3)	24-vertex cubic	0
S_4	(2, 3, 4)	24-vertex cubic	0
S_4	(3, 3, 3)	24-vertex cubic	0
S_4	(3, 4, 4)	24-vertex cubic	0
A_5	(2, 3, 5)	60-vertex cubic, girth 4	1
A_5	(2, 5, 5)	60-vertex cubic, girth 4	1
A_5	(3, 3, 5)	60-vertex cubic, girth 6	0
A_5	(3, 5, 5)	60-vertex cubic, girth 6	0
A_5	(5, 5, 5)	60-vertex cubic, girth 9	0

Table 3: Free-orbit templates and their Node-Kayles values for the polyhedral groups. These are free-orbit values only; the computation and its cross-checks are described in Appendix B.

Remark 6.1. In A_5 the two value-one classes are exactly those containing a commuting generator pair, equivalently a pair product of order two. This is an observation about the displayed table, not a criterion used in the computation and not a general theorem: the analogous (2, 3, 3) and (2, 3, 4) classes of S_4 both have value zero.

Together with the Klein-four and dihedral results above, Table 3 completes the free-orbit row t_1 for every tame small-subgroup type generated by a legal involution triple.

7 Arithmetic synthesis

This section converts the template tables of Sections 4–6 into closed formulas over $\mathbb{F}_q$. Throughout, $p \nmid |G|$.

7.1 Closed values for the three-point family

Assume $p \nmid 4n$.

Split torus

Suppose $2n \mid q-1$ and choose $r : t \mapsto \omega t$, $s : t \mapsto t^{-1}$ with ω of order $2n$. The rotation torus fixes $0, \infty$, so $\varepsilon = 1$. The two reflection classes are represented by $t \mapsto t^{-1}$ and $t \mapsto \omega t^{-1}$. The first is split; the second is split exactly when ω is a square, equivalently when $4n \mid q-1$. Put

$$h_- = \frac{q-1}{2n}, \quad \delta_- = \mathbf{1}_{4n \mid q-1}.$$

Then

$$a_0 = 1, \quad a_1 = \delta_-, \quad f = \frac{h_- - 1 - \delta_-}{2}. \quad (31)$$

For odd d ,

$$\mathcal{G}(R_{T_d}) = \left(\frac{h_- - 1 - \delta_-}{2} \bmod 2 \right) \oplus (\mathbf{1}_{2 \mid n}(1 - \delta_-)), \quad (32)$$

and for even d ,

$$\mathcal{G}(R_{T_d}) = 1 - \delta_-. \quad (33)$$

Nonsplit torus

Suppose $2n \mid q+1$. Model the projective line by the norm-one group $U = \{x \in \mathbb{F}_{q^2}^\times : x^{q+1} = 1\}$ and choose $r : x \mapsto \omega x$, $s : x \mapsto x^{-1}$ with $\omega \in U$ of order $2n$. The rotation torus has no rational fixed points, so $\varepsilon = 0$. The two reflection classes have fixed-point equations $x^2 = 1$ and $x^2 = \omega$, so the first is split and the second is split exactly when $4n \mid q+1$. Put

$$h_+ = \frac{q+1}{2n}, \quad \delta_+ = \mathbf{1}_{4n \mid q+1}.$$

Then

$$a_0 = 1, \quad a_1 = \delta_+, \quad f = \frac{h_+ - 1 - \delta_+}{2}, \quad (34)$$

and

$$\mathcal{G}(R_{T_d}) = \left(\frac{h_+ - 1 - \delta_+}{2} \bmod 2 \right) \oplus (\mathbf{1}_{2 \mid n}(1 - \delta_+)) \quad (35)$$

for odd d , while for even d ,

$$\mathcal{G}(R_{T_d}) = 1 - \delta_+. \quad (36)$$

Corollary 7.1 (Explicit P-position congruences). *In either torus family write $h = (q-1)/2n$ in the split case and $h = (q+1)/2n$ in the nonsplit case. Then the conic-only residual is a P-position exactly under the following congruences:*

triple type	P exactly when
d odd, n odd	$h \equiv 1, 2 \pmod{4}$
d odd, n even	$h \equiv 2, 3 \pmod{4}$
d even (n odd)	$h \equiv 0, 2 \pmod{4}$

(37)

Proof. In both torus families $\delta = 1$ exactly when h is even. Substitution into (32)–(36), followed by the four cases $h \bmod 4$, gives the table. $\square$

[C307 owed] **Correctness hazard 1 — blocking.** For even h a second D_{4n} conjugacy class with all-nonsplit reflections ($t = 0$) exists, first at $q = 7$, D_8 acting freely with board M_8 and value 1 against the boxed 0. C281 established that equations (32)–(36) and Corollary 7.1 require a t -case split, and that for odd d exactly one of the two classes is an N-position. C307 must apply that correction here before any polishing; the formulas above are migrated verbatim from the source manuscript and are known to be incomplete as stated. The §14 pair value formula survives unchanged ($t \equiv 1 + \delta \bmod 2$).

For the even- d triple the criterion has the geometric form

$$R_{T_d} \text{ is P} \iff \text{both reflection classes are split} \iff 4n \mid q \mp 1, \quad (38)$$

where the upper sign is the split-torus family and the lower sign the nonsplit-torus family. For fixed n these formulas are periodic in q with period dividing $8n$ within each torus family.

7.2 Closed values for the two-point family

Theorem 7.2 (Closed Grundy value of the pair family). *Let $G \cong D_{2m}$ act tamely on $\mathbb{P}^1(q)$ through a split torus ($m \mid q-1$) or a nonsplit torus ($m \mid q+1$), and put $\delta = \mathbf{1}_{2m \mid q \mp 1}$ with the upper sign for the split family and the lower sign for the nonsplit family. Then*

$$\mathcal{G}(R_T) = \begin{cases} 0, & m \text{ odd,} \\ (1 - \delta) \mathcal{G}(P_m), & m \text{ even.} \end{cases} \quad (39)$$

Proof. By (14) the value is $(\rho \bmod 2) \mathcal{G}(P_m)$, so it suffices to compute the parity of the number ρ of reflection orbits.

Odd m . The reflections form a single conjugacy class, so they are all split or all nonsplit. If nonsplit, no reflection has a rational fixed point and $\rho = 0$. If split, each of the m reflections has two rational fixed points, and a conic point is fixed by at most one reflection because a tame point stabilizer is cyclic; so the reflection fixed points form a set F of exactly $2m$ points. Every point of F has stabilizer a reflection subgroup of order two, hence lies in an orbit of size $|G|/2 = m$. Therefore F splits into $2m/m = 2$ orbits and $\rho = 2$. In both cases ρ is even and the value is 0.

Even $m = 2n$. Now the central involution $z = r^n$ exists. It commutes with every reflection σ , so it permutes the two-point set $\text{Fix}(\sigma)$. The stabilizer of a reflection fixed point is exactly $\langle \sigma \rangle$, which does not contain the rotation z ; hence z fixes neither point of $\text{Fix}(\sigma)$ and therefore swaps them. So for each reflection class the two rational fixed points, when they exist, lie in a single orbit, and each split reflection class contributes exactly one reflection orbit. In the split-torus model $r : t \mapsto \omega t$, $s : t \mapsto t^{-1}$ with ω of order m , the class $\langle s \rangle$ fixes $t = \pm 1$ and is always split, while $\langle sr \rangle$ fixes the solutions of $t^2 = \omega^{-1}$, which are rational exactly when ω^{-1} is a square, that is when $2m \mid q - 1$; the nonsplit-torus model gives the same statement with $q + 1$. Therefore $\rho = 1 + \delta$, $(\rho \bmod 2) = 1 - \delta$, and the value is $(1 - \delta) \mathcal{G}(P_m)$. $\square$

The even case is the two-generator counterpart of the three-generator group D_{4n} with $m = 2n$; the reflection-splitting condition $2m \mid q \mp 1$ is the condition $4n \mid q \mp 1$ of (38).

Corollary 7.3 (P-positions of the pair family). *The two-point residual R_T is a P-position exactly when*

$$m \text{ is odd, } \quad \text{or} \quad \mathcal{G}(P_m) = 0, \quad \text{or} \quad 2m \mid q \mp 1. \quad (40)$$

Proof. Immediate from Theorem 7.2. $\square$

7.3 Density among admissible prime fields

For fixed (G, T) the projective-line value is determined by $q \bmod 2|G|$ together with the bounded list of exceptional orbit indicators. This follows from the orbit equation

$$q + 1 = |G|f + \sum_K a_K [G : K], \quad (41)$$

which determines $f \bmod 2$. For the dihedral family the exceptional indicators are controlled by $4n \mid q \mp 1$, so period $8n$ suffices.

Theorem 7.4 (Density one half). *Fix $n \geq 2$, a legal triple type, and either the split or the nonsplit torus family. Among admissible prime fields, the relative Dirichlet density of P-positions is $\frac{1}{2}$, and the relative density of N-positions is also $\frac{1}{2}$.*

Proof. Let $h = (q - 1)/2n$ in the split case and $h = (q + 1)/2n$ in the nonsplit case. The four possibilities $h \bmod 4$ give four admissible reduced residue classes modulo $8n$: the corresponding prime candidate is $q = 2nh \pm 1$, which is odd, and any common divisor with n would divide ± 1 , so $\gcd(q, 8n) = 1$ for all four values of $h \bmod 4$. Corollary 7.1 selects exactly two of these four classes in every triple type. The prime number theorem for arithmetic progressions, equivalently its standard Dirichlet-density consequence [7], gives the same Dirichlet density to each reduced residue class. Relative to primes in the chosen torus family the four classes therefore each have density $1/4$, proving the claim. $\square$

[C307/C308 owed] Theorem 7.4 is one of the paper’s two Lean-anchored headline results and must keep that status (Fable’s change 1). C307 attaches the C278 boundary sentence: the density- $\frac{1}{2}$ kernel certificate is conditional on exactly one quarantined equidistribution axiom (`primes_equidistribute`), audited as [`propext`, `Classical.choice`, `Quot.sound`, `primes_equidistribute`]. The paper must not imply an axiom-free Lean proof. C307 also re-derives the statement against the corrected Corollary 7.1.

Corollary 7.5 (Density for the pair family). *Fix $m \geq 3$ and a torus family. Among admissible prime fields the relative Dirichlet density of P-positions for the two-point game is*

$$\begin{cases} 1, & m \text{ odd, or } m \text{ even with } \mathcal{G}(P_m) = 0, \\ \frac{1}{2}, & m \text{ even with } \mathcal{G}(P_m) \neq 0. \end{cases} \quad (42)$$

Proof. An odd m or a Dawson zero $\mathcal{G}(P_m) = 0$ makes every admissible field a P-position by Corollary 7.3. Otherwise, with $h = (q \mp 1)/m$, the indicator $\delta = \mathbf{1}_{2m|q \mp 1}$ equals $\mathbf{1}_{2|h}$; the two parities of h correspond to two reduced residue classes modulo $2m$, each of equal Dirichlet density among primes, so P-positions have relative density $\frac{1}{2}$. $\square$

7.4 Worked examples

Example 7.6 (D_{12}). Take $n = 3$, so $G = D_{12}$ has rotation order six, and there are two legal triple types.

For odd d , each free orbit contributes $\mathcal{G}(M_{12}) = 1$ while every reflection orbit contributes $\mathcal{G}(L_2) = 0$; the value is simply $f \bmod 2$.

For even d , each free orbit is the prism $C_6 \square K_2$ and contributes zero, while each split reflection class contributes a ladder L_3 or L_1 , both of value one; the total value is $a_0 \oplus a_1$. In the split-torus family $6 \mid q - 1$, the second reflection class is split exactly when $12 \mid q - 1$, so the even- d position is P exactly when $12 \mid q - 1$.

[C307 owed] Fable’s answer (6): keep the D_{12} example and add one contrasting polyhedral example, drawn from the A_5 rows, rather than replacing it. That example depends on the C290 congruence laws and is therefore owed by C307, not by the structural rebuild.

8 Realization, boundaries, and outlook

8.1 Converse realization

Theorem 8.1 (Realization, three-point family). *For the Klein-four boundary, K_4 occurs over infinitely many prime fields as a transitive residual template of a legal tame conic-involution triple. For every $n \geq 2$, the graphs*

$$M_{4n}, \quad L_{n-1} \quad (43)$$

and the completely deleted rotation-torus template occur over infinitely many prime fields as transitive residual templates of legal tame dihedral conic-involution triples. If $n \geq 3$ is odd, the same is true of

$$C_{2n} \square K_2, \quad L_n, \quad L_{n-2}. \quad (44)$$

Proof. The abstract coset calculations of Section 5 produce the listed graph whenever the corresponding orbit type occurs.

For the Klein-four assertion, take primes $q \equiv 3 \pmod{4}$ and the standard commuting triple

$$t \mapsto -t, \quad t \mapsto a/t, \quad t \mapsto -a/t$$

with a a square. Its centres form a legal self-polar triangle. Exactly two of the three involutions are split, so Theorem 5.1 gives $(q - 3)/4$ free K_4 -orbits, which is positive for $q \geq 7$. Dirichlet's theorem supplies infinitely many such primes.

Choose primes $q \equiv 1 \pmod{2n}$ for split-torus realizations or $q \equiv -1 \pmod{2n}$ for nonsplit-torus realizations; stronger congruences modulo $4n$ realize both reflection classes, and Dirichlet's theorem supplies infinitely many such primes. The triples T_d are legal by Section 5. For sufficiently large primes in any chosen class, equation (30) gives $f > 0$, so the free template occurs. Reflection and rotation-torus templates occur under the splitting conditions of Section 7. $\square$

Theorem 8.2 (Realization, two-point family). *For every $m \geq 3$, the cycle C_{2m} and the path P_m occur as transitive residual templates of legal tame conic-involution pairs over infinitely many prime fields.*

Proof. Choosing $q \equiv -1 \pmod{m}$ (nonsplit) or $q \equiv 1 \pmod{m}$ (split) realizes a D_{2m} -action with $m = \text{ord}(\sigma_x \sigma_y)$. For large such primes the orbit count gives a free orbit, hence C_{2m} , while a congruence modulo $2m$ that makes a reflection class split produces the reflection orbit P_m . Dirichlet's theorem supplies infinitely many primes in each class. $\square$

8.2 The wild boundary

Every result above assumes the tame hypothesis $p \nmid 2m$. When $p \mid 2m$ — equivalently, since q is odd, $p \mid m$ — the classification changes character completely, for one structural reason. A cyclic subgroup of $\text{PGL}_2(q)$ whose order is divisible by p has order exactly p , because a nonsemisimple element of $\text{PGL}_2(q)$ is unipotent of order p . Hence the rotation $r = \sigma_x \sigma_y$ can never have order $2p, 3p, \dots$, and the only wild dihedral group that arises is D_{2p} , with r a transvection.

Three consequences follow. First, r has a single fixed point on $\mathbb{P}^1(q)$ instead of the tame 0 or 2, so the deleted set $D_S = \text{Fix}(r)$ is a single point and the even orbit-deletion pattern of Section 4 fails. Second, that fixed point is fixed by the whole group: D_{2p} lies in a Borel subgroup and acts reducibly on $\mathbb{P}^1(q)$, so there is no regular orbit and no cycle template C_{2p} ; the action splits into the deleted fixed point and one orbit of size p with stabilizer of order 2. The residual is therefore a single path,

$$R_S \cong P_p, \quad \mathcal{G}(R_S) = \mathcal{G}(P_p) = \text{A002187}(p).$$

Third, because $m = p$ is odd, this directly contradicts the tame law that odd-order dihedral games are P-positions (Theorem 7.2): the wild game D_{2p} is generically an N-position. The smallest odd nontrivial case $p = 5$ gives D_{10} with r unipotent, D_S a single point, residual P_5 , and $\mathcal{G} = \text{A002187}(5) = 3 \neq 0$ — an N-position, whereas every tame D_{10} is a P-position. Legal triples cannot be wild at all: a wild rotation is unipotent of odd order and has no central involution, so no legal generating triple exists.

[C307 owed] Fable's change 3: this wild result is the paper's sharpest boundary marker and should be presented as such rather than as an aside. Whether to promote it from remark to a numbered lemma is a user scope call; C283 judged it a short lemma and the default is the remark as landed.

8.3 Boundaries of the classification

Three boundaries should be distinguished sharply.

- (1) **Completed here.** The tame dihedral configuration families of every rotation order and both configuration sizes, together with the finite polyhedral boundary of Section 6.

- (2) **Wild polyhedral characteristic.** Subsection 8.2 settles the wild dihedral pair; the wild polyhedral case is not treated here.
- (3) **Growing full or subfield groups.** In the motivating projective-cap game, a generic fourth move typically generates the full group $\mathrm{PSL}_2(q)$ or $\mathrm{PGL}_2(q)$. Evaluating or controlling those growing residual templates is a separate problem, and no claim about it is made here.

8.4 Discussion

The classification of legal dihedral configurations is complete for every rotation order and both configuration sizes, and has a simple conceptual form. A legal tame dihedral conic configuration produces only K_4 's, Möbius ladders, prisms, ordinary ladders, cycles, paths, and completely deleted exceptional orbits. Its conic-only Node-Kayles value is an explicit congruence function of the field and the embedded configuration. Three selected points give the even-rotation groups D_{4n} and values in $\{0, 1\}$; two selected points give every D_{2m} , with Dawson's-chess values, and the odd-order groups are always P-positions.

The fixed-point-deleted Schreier construction is what separates geometry from game theory. The abstract pair (G, T) determines a finite table of transitive template numbers; the ambient geometry determines only the parity vector of orbit multiplicities. The three-point and two-point families share this reduction and differ only in that table: ladders and Möbius ladders for the former, paths and cycles for the latter.

[C311 owed] The Discussion is rewritten last, after the theorem boundary and the final title are fixed (runbook migration table, current §15 row).

A Evidence and validation map

[C307/C308 owed] This appendix is a structural placeholder. C307 populates it with the C281 dihedral census (all tame legal pairs and triples over prime $q \leq 23$) and the C288 polyhedral embedding census (odd prime powers $q \leq 101$), together with the polyhedral edge lists and replay counts displaced from Section 6. C308 adds the mixed-verification claim-route map, the Lean adequacy statement, and the provenance boundary. C309 adds the regeneration commands and manifest hashes. Censuses are validation evidence, not narrative spine.

The path and cycle Node-Kayles values used in Section 4, the even-index property of the Dawson zeros underlying Lemma 4.3, and the full two-point classification of (10), (14) and (39) were verified end to end by an exhaustive conic simulation over the prime fields $q \in \{5, 7, 11, 13, 17, 19, 23\}$: for every legal pair the fixed-point-deleted residual graph was built directly and its Node-Kayles value matched the orbit formula, with no discrepancy.

B Regular polyhedral templates: exact computation

The computation behind Table 3 uses memoized Sprague-Grundy recursion with connected-component xor and canonical masks under graph automorphisms. Left multiplication supplies $|G|$ automorphisms in every class. The symmetric $(5, 5, 5)$ triple also admits the colour-permuting S_3 , giving a 360-element canonicalization group.

The four S_4 values were cross-checked by two independent solvers. The five A_5 values were subsequently independently cross-checked as well: a separate solver using a different canonicalization reproduces all five, and an independent graph-isomorphism canonicalization confirms that the automorphism group used equals the full automorphism group of each Cayley graph.

These are free-orbit values only. In particular the uniform value zero for regular S_4 orbits does not evaluate a generic fourth-move residual whose generated group has grown to full $\text{PSL}_2(q)$ or $\text{PGL}_2(q)$.

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